

CUSTOMER REVIEW INTELLIGENCE: LLM-ASSISTED NLP FOR BUSINESS INSIGHT EXTRACTION

Technical Case Study

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1. Executive Summary

This project explores how large-scale customer review data can be transformed into **structured operational insight** using Natural Language Processing (NLP), transformer-based topic modelling, emotion analysis, and LLM-assisted thematic abstraction.

The objective was not simply to measure sentiment, but to design an **end-to-end insight generation pipeline** capable of identifying recurring customer pain points, uncovering emotional drivers behind complaints, and translating unstructured feedback into actionable business recommendations.

Multiple analytical approaches were compared — including classical topic modelling, embedding-driven clustering, emotion-aware filtering, and local LLM-based summarisation — to determine which combination produced the most interpretable and decision-useful outputs.

The strongest workflow combined:

BERTopic clustering + emotion filtering + LLM-assisted topic abstraction

This produced clearer themes, richer interpretation, and stronger recommendation quality than traditional NLP methods alone.

2. Problem Context

Customer reviews contain valuable business intelligence, but extracting consistent insight at scale is difficult because:

- reviews are unstructured text
- customers describe similar issues differently
- complaint themes often overlap
- sentiment alone lacks operational context
- recurring issues vary across branches / locations
- manual review is time-intensive and inconsistent

The goal of this project was to build a scalable NLP workflow that could:

- identify recurring complaint themes
- compare topic modelling methods
- analyse emotional patterns in customer feedback
- segment issues geographically
- generate recommendation-oriented insight outputs

3. Dataset & Analytical Scope

The project analysed customer reviews collected from:

- Google Reviews
- Trustpilot reviews

The analysis focused particularly on:

- negative reviews
- recurring complaint clusters
- branch/location-level issue patterns
- operational pain points with business actionability

This framing shifted the work from generic sentiment analysis toward **decision-support analytics**.

4. Data Preparation & Exploratory Analysis

Text preprocessing included:

- regex-based cleaning
- lowercasing / normalisation
- tokenisation
- stopword removal
- punctuation filtering
- NLP preprocessing using NLTK

Exploratory text analysis included:

- word frequency distributions
- keyword concentration analysis
- phrase inspection
- WordCloud visualisation
- branch-level review comparisons

This helped establish early signals before formal modelling.

5. Topic Modelling Approach Comparison

A central goal of the project was to compare alternative topic extraction strategies.

LDA Baseline

Latent Dirichlet Allocation was used as a classical benchmark model.

Strengths:

- simple probabilistic topic structure
- interpretable baseline comparison

Limitations:

- weaker semantic grouping
- reduced coherence in overlapping themes

BERTopic

BERTopic was implemented as the primary embedding-driven topic modelling framework.

Advantages:

- semantically coherent topic clustering
- flexible topic granularity
- stronger interpretability
- better handling of nuanced customer complaints

Visual diagnostics included:

- Intertopic Distance Map
- topic bar charts
- topic similarity heatmaps

This significantly improved thematic clarity compared with LDA.

6. Emotion-Aware Insight Layer

Emotion classification was performed using:

bhadresh-savani/bert-base-uncased-emotion

This expanded analysis beyond polarity (positive/negative) into richer behavioural signals such as:

- anger
- frustration
- disappointment
- satisfaction
- excitement

Emotion-aware filtering improved prioritisation of operationally meaningful complaint clusters.

This helped distinguish:

minor dissatisfaction vs high-friction operational failure points

7. LLM-Assisted Topic Abstraction

To improve theme naming and recommendation synthesis, local LLM workflows were explored in Google Colab.

Models evaluated included:

- FLAN-T5
- Falcon-7B
- Phi-4-mini
- **Qwen2.5-3B-Instruct** (selected)

Qwen was chosen because it offered:

- stronger prompt adherence
- practical inference efficiency
- better topic summarisation quality
- easier Colab execution

LLM outputs were used for:

- clearer topic labelling
- abstraction of complaint clusters
- summarisation of customer pain points
- business recommendation generation

8. Key Findings

The strongest analytical workflow combined:

BERTopic + Emotion Analysis + LLM Abstraction

This produced:

- clearer theme separation
- richer contextual interpretation
- better operational relevance
- more actionable recommendations
- stronger business readability

A major insight from the project was:

Combining classical NLP, embedding-based clustering, and LLM synthesis creates stronger decision-support insight than using any single method independently.

9. My Contribution

This was an end-to-end individual project where I independently led:

- data preparation and cleaning
- exploratory text analysis
- topic modelling experimentation
- BERTopic implementation
- emotion analysis pipeline design
- local LLM experimentation and prompting
- comparative evaluation across approaches
- insight synthesis and business recommendation generation
- visualisation and reporting

10. Limitations

Current limitations include:

- review-source bias
- dependence on preprocessing quality
- prompt sensitivity in LLM summarisation
- branch-level imbalance in review volume
- computational constraints for larger local models

11. Future Extensions

Potential future enhancements:

- retrieval-based review search
- aspect-based sentiment modelling
- multilingual review processing
- dashboard deployment for business teams
- automated recommendation generation pipelines
- trend monitoring over time

12. Key Learnings

This project strengthened understanding of:

- practical NLP pipeline design
- transformer-based topic modelling
- emotion-aware text analytics
- prompt engineering for summarisation
- comparative model evaluation
- translating text analytics into decision-support insight

A key takeaway:

The real value of NLP lies not in finding themes — but in converting customer voice into structured, actionable intelligence that supports business decisions.